ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** variety of life	Diversity.
2	**(d)** Porifera	Pores.
3	**(b)** Reptilia	Scales.
4	**(d)** similarities and differences	Characters.
5	**(c)** Cnidaria	Stinging cells.
6	**(a)** Aves	Feathers.
7	**(b)** binomial nomenclature	Naming.
8	**(d)** flatworm	Flat.
9	**(c)** Mammalia	Hair and milk.
10	**(d)** species	Second word.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grouping of organisms based on similarities (2).	2
12	Group of organisms that can interbreed (2).	2
13	Porifera (1); Arthropoda (1).	1+1
14	Science of naming, describing and classifying organisms (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell type (1); Body organisation (1); Nutrition (1).	1+1+1
16	No vascular tissue (1); Need water (1); Small (1).	1+1+1
17	Gills (1); Fins (1); Scales (1).	1+1+1
	OR: Kingdom, phylum, class (1); Order, family (1); Genus, species (1).	1+1+1
18	Vascular tissue (1); Spores (1); No seeds (1).	1+1+1
	OR: Cotyledon (1); Venation (1); Roots (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Monera (1); Protista (1); Fungi (1); Plantae (1); Animalia (1).	1+1+1+1+1
	OR: Cotyledons (1); Roots (1); Leaves (1); Flowers (1); Examples (1).	1+1+1+1+1
20	Algae (1); Bryophyta (1); Pteridophyta (1); Gymnosperms (1); Angiosperms (1).	1+1+1+1+1
	OR: Study (1); Identification (1); Relationship (1); Evolution (1); Biodiversity (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** roundworm	Round.
2	**(c)** mammary glands	Feed milk.
3	**(a)** prokaryotes	Monera.
4	**(b)** annelid	Segmented.
5	**(d)** feathers	Wings.
6	**(b)** fungi	Fungi.
7	**(a)** arthropod	Insect.
8	**(b)** dry scaly skin	Land.
9	**(d)** angiosperm	Flowering.
10	**(c)** mollusc	Shell.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Two-word naming system (2).	2
12	Pisces, Amphibia, Reptilia (1); Aves, Mammalia (1).	1+1
13	R. H. Whittaker (2).	2
14	Monocot (1); Dicot (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Homo sapiens (1); Mangifera indica (1); Panthera leo (1).	1+1+1
16	Hair (1); Mammary glands (1); Give birth (1).	1+1+1
17	Jointed legs (1); Exoskeleton (1); Segmented body (1).	1+1+1
	OR: Monera: bacteria (1); Fungi: yeast (1); Plantae: mango (1).	1+1+1
18	Feathers (1); Beak (1); Wings (1).	1+1+1
	OR: Soft body (1); Shell (1); Muscular foot (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Porifera (1); Cnidaria (1); Worms (1); Arthropoda, Mollusca (1); Chordata (1).	1+1+1+1+1
	OR: Bacteria (1); Fungi (1); Uses (1); Harm (1); Examples (1).	1+1+1+1+1
20	Pisces (1); Amphibia (1); Reptilia (1); Aves (1); Mammalia (1).	1+1+1+1+1
	OR: Features (1); Jointed legs (1); Examples (1); Importance (1); Harm (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Patterns in Life: Diversity and Classification** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** fins	Swim.
2	**(b)** gymnosperm	Naked seeds.
3	**(a)** echinoderm	Spiny skin.
4	**(a)** monocot	One cotyledon.
5	**(c)** pteridophytes	Vascular.
6	**(a)** Pisces	Gills.
7	**(c)** dicot	Two cotyledons.
8	**(d)** bryophyte	Non-vascular.
9	**(d)** Amphibia	Water and land.
10	**(a)** species	Basic unit.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heterotrophic eukaryotes with cell walls of chitin (2).	2
12	Monera, Protista, Fungi (1); Plantae, Animalia (1).	1+1
13	Pine (1); Cycas (1).	1+1
14	Simple plants with a thallus body, mostly aquatic (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Flowers (1); Seeds in fruits (1); Vascular tissue (1).	1+1+1
16	Scales (1); Lay eggs (1); Cold-blooded (1).	1+1+1
17	Bread making (1); Medicines (1); Decomposition (1).	1+1+1
	OR: Naked seeds (1); Cones (1); Woody (1).	1+1+1
18	Moist skin (1); Live in water and land (1); Lay eggs in water (1).	1+1+1
	OR: Curd (1); Nitrogen fixation (1); Decomposers (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Genus (1); Species (1); Capital and small (1); Italics (1); Example (1).	1+1+1+1+1
	OR: Birds features (1); Mammal features (1); Examples (1); Warm-blooded (1); Differences (1).	1+1+1+1+1
20	Kingdom (1); Phylum (1); Class and order (1); Family (1); Genus and species (1).	1+1+1+1+1
	OR: Reptile features (1); Amphibian features (1); Examples (1); Habitat (1); Differences (1).	1+1+1+1+1

Mistakes to avoid:

- Writing scientific names wrongly: *homo Sapiens* or *Homo Sapiens*; the genus has a capital letter, the species a small letter, and both are italicised or underlined separately.
- Treating the kingdom as the smallest group and the species as the largest; the hierarchy runs from kingdom (largest) to species (smallest).
- Putting fungi (mushroom, yeast) in the plant kingdom; they lack chlorophyll and have chitin in their cell wall.
- Mixing up gymnosperms (naked seeds) and angiosperms (seeds enclosed in fruits), or monocots and dicots (one or two cotyledons).
- Saying that all vertebrates are warm-blooded, or that bats (mammals) are birds because they fly.
- Mixing up the phyla: spiders and insects are arthropods, earthworms are annelids, and roundworms are nematodes; also confusing 'Annelida' (segmented) with 'Nematoda' (unsegmented).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.